Equivariant orbit aggregation for paired cuts

Human Compression project: L044 continuation

Let $H = \langle s_0, \dots, s_{r-1} \rangle$ act on a finite set U, with each s_i an involution or the identity and $[s_i, s_j] = 1$ for $|i-j| > 1$. Write $O_1, \dots, O_c$ for its actual orbits. The full local omission on O_a is

$$H_{a,j} = \langle s_i|_{O_a} : i \neq j \rangle.$$

Assign pairwise disjoint colour sets $F_1, \dots, F_c$ to these orbits. For every $j \in F_a$, assume that $s_j|_{O_a}$ has at least two edges whose endpoints belong to different $H_{a,j}$-orbits. Set $d_a = |F_a| - |O_a|/2$. The accepted selected-colour theorem, applied to the transitive full action on each orbit, gives $d_a \leq 0$. The assigned sets need not exhaust all colours unless explicitly stated.

Lemma 1 (Transport and aggregation)**.** *Suppose $\phi : O_a \longrightarrow O_b$ is a bijection satisfying $\phi(v^{s_i}) = \phi(v)^{s_i}$ for every i and $v \in O_a$. Put $m = |O_a| = |O_b|$, with $a \neq b$. Then*

$$|F_a| + |F_b| \leq \lfloor m/2 \rfloor, \qquad d_a + d_b \leq -\lceil m/2 \rceil. \tag{1}$$

If $m = 3$, then $F_a = F_b = \varnothing$ and $d_a + d_b = -3$.

Proof. The bijection conjugates each marked restricted generator and hence every full local omission $H_{a,j}$ to $H_{b,j}$. It carries their orbit partitions to one another. For each colour in F_b, pull its two eligible edges back to O_a. Together with those already chosen for F_a, these give two eligible edges for every colour in $F_a \cup F_b$ on O_a. Disjointness gives $|F_a \cup F_b| = |F_a| + |F_b|$; without that hypothesis this equality would not follow. The selected-colour theorem now bounds the union by $\lfloor m/2 \rfloor$, proving (1). An involution on three points has at most one edge, so no colour can be assigned to an orbit of size three under the standing hypothesis. □

Theorem 2 (An exchanged odd orbit pair)**.** *Suppose an involution δ centralizes H and exchanges two distinct nontrivial H-orbits O_a, O_b of odd size. Then*

$$\sum_{s=1}^{c} d_s \leq -3. \tag{2}$$

More explicitly, if their common size is $m \geq 5$, their combined deficit is at most $-(m+1)/2$; for $m = 3$ it is -3.

Proof. Centralization makes $\delta|_{O_a}$ the marked equivariant bijection of Lemma 1. Nontriviality and odd size give $m \geq 3$. The lemma proves the stated deficit in both ranges; all remaining d_s are nonpositive. Summing proves (2). On the exchanged pair δ is a product of exactly m transpositions, so this is precisely an odd exchanged-orbit contribution to its sign. An exchanged pair of singleton orbits would instead contribute only -1 and is not covered by the theorem. □

Corollary 3 (Exact numerical tail interface)**.** *Suppose the standing action is the restriction of a string tail $\langle \rho_{t+1}, \dots, \rho_{r-1} \rangle$ to its support U. Assume the assigned sets partition all $r-1-t$ tail indices, the full local two-edge hypotheses hold, and Theorem 2 applies. Then*

$$r - 1 - t \leq |U|/2 - 3. \tag{3}$$

In the two source head configurations, $t = 1$ with $|U| \leq n-2$, or $t = 4$ with $|U| \leq n-7$, this implies $r \leq (n-3)/2$.

Proof. Partitioning gives $r - 1 - t = \sum_s |F_s|$, so (2) gives (3). The first configuration yields $r \leq (n-4)/2$ and the second $r \leq (n-3)/2$, as required. No assertion about a particular source assignment supplies the local eligibility hypotheses here. □

Source interface and attribution. Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, physical page 42, Proposition 5.31, case (1), uses an odd pair of exchanged nontrivial tail orbits. The orbit assignment and deficit framework are borrowed from its Propositions 5.27 and 5.31. The present argument replaces the maximal-empty-assignment step by a direct marked-equivariance aggregation, with a bound valid for every odd orbit size. The full accepted paired-cut and equality proofs remain paid through the selected-colour theorem; this is not an independent reproof of those results. No projected intersection property is used.

Scope. This proves the algebraic deficit route in case (1), conditional on the exact local hypotheses above. Full local omissions are the images of the tail omissions on each orbit; they are not silently identified with omissions of an ambient larger group or of the selected-colour subgroup. No assertion is made that the source's other orbit configurations meet the two-edge hypothesis. The internal-orbit cases (2)–(4), their equality rigidity and the remaining source tables are not resolved here. There is no new finite computation, formal theorem, whole-original recovery, conjecture solution or priority claim.